

Project Title
Pathogen Genomic Surveillance & Outbreak-Tracing System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Pathogen Genomic Surveillance & Outbreak-Tracing System

During infectious disease outbreaks, public health laboratories generate large volumes of pathogen genome sequences that must be rapidly analyzed to understand transmission dynamics and guide containment efforts. Sequencing every sample is often infeasible due to budget and turnaround-time constraints, making it critical to identify which samples carry the most epidemiological value before committing resources to full sequencing.

This project proposes an integrated genomic surveillance engine that indexes pathogen sequences with fast retrieval, computes pairwise relatedness via alignment-based and alignment-free (k-mer/sketch) similarity, and detects mutations between related genomes, flagging high-frequency mutation positions. From relatedness scores, the system constructs a transmission network, partitions it into outbreak clusters, and identifies critical transmission links, including bridge cases and potential superspreading events, whose removal would significantly disrupt inferred transmission pathways. A contact-spread simulator lets users test intervention scenarios by isolating samples or breaking links and observing the resulting change in network connectivity.

To address budget constraints, a sample-prioritization module formulates sequencing selection as a coverage optimization problem, recommending the minimum sample set that maximizes genomic and network coverage. The system scales incrementally as new sequences arrive, ranks clusters by structural growth risk, and surfaces results — relatedness matrices, cluster maps, mutation timelines, and prioritized sample lists — through an interactive dashboard with threshold-based alerts for public-health decision-makers

Signature of the Guide